

THE PRACTICAL GUIDE TO

Full Closed Loop

* * *

And a Comprehensive Exploration of autoISF

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Restructured and edited for clarity

A U T H O R

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This edition used AI to restructure and edit Bernie's original e-book for improved clarity.

The original materials are available at:

github.com/Bernie4375/FCL

Important Note: This edition may not be thoroughly reviewed or updated by Bernie or other key authors of original autoISF material. Use of this edition is at your own risk.

Please cross-check any important information against the original material, and be sure to alert the community or original authors of anything that may require correction.

To contact Bernie or to access a wealth of additional resources and information, please visit the Full-Closed-Looping community Discord channel:

discord.gg/EYxzq7GMX

* PhD in Technical Chemistry. Focus of research: Calculating permanently disturbed and re-establishing equilibria in chemical reactors.

Having lived with T1D for decades, trying to understand what's going on in my little "reactor" (my stomach and body cells) is second nature, though it remains a struggle 😊 ...

Table of Contents

The book is organized in seven Parts that map the journey from “should I do this?” through setup, day-to-day operation, and long-term monitoring. The Parts are cumulative — each assumes you’ve done the work of the ones before it.

FRONT MATTER

Disclaimer · Please Read This First

Preface · How to Use This Book

PART I — GETTING ORIENTED

What this is, whether it’s for you, and what the journey looks like.

Introduction · What Is autoISF Full Closed Loop?

Beginner’s Roadmap · The Journey to Full Closed Loop

PART II — FOUNDATIONS

What must be true before you start, and how the algorithm actually works.

Chapter 1 · Prerequisites for Full Closed Loop

Chapter 2 · How autoISF Works

PART III — CORE SETUP

The multi-week project at the heart of the book — widening safety limits, installing the new guardrails, and tuning ISF weights to your meals.

Chapter 3 · General FCL Settings

Chapter 4 · Meals: Setting Your ISF Weights

PART IV — DAY-TO-DAY OPERATIONS

Nudging the loop when it needs nudging — snacks, exercise, sick days, and everything else that isn't a normal meal.

Chapter 5 · Modulating Loop Aggressiveness

Chapter 6 · Exercise and Activity

PART V — ALTERNATIVE MODES

Gentler paths when full autoISF FCL isn't the right fit — hybrid with Meal Announcement, simpler FCL via AAPS Automations, and the broader landscape.

Chapter 7 · Advanced HCL with Meal Announcement

Chapter 8 · Other Avenues to Full Closed Loop

PART VI — MONITORING & TROUBLESHOOTING

Keeping the loop healthy once it's running, and fixing it when something goes wrong.

Chapter 9 · Performance Monitoring

Chapter 10 · Troubleshooting

PART VII — TOOLS

Optional but powerful: the Emulator for replaying and analyzing your loop's decisions.

Chapter 11 · The Emulator on Your PC

Chapter 12 · The Emulator on Your Phone

BACK MATTER

Final Remark

Appendix A · Glossary

Appendix B · Case Studies Index

DISCLAIMER

Please Read This First



This book is not medical advice.

The authors are people living with Type 1 diabetes (or parents of children with T1D). We are not medical professionals. What you will read is our limited understanding, based on personal experience, shared openly to contribute to a growing body of community knowledge.

autoISF is an early-stage development tool. It is not a medical product. Good safety features exist, but they only work as well as the person using them understands them. The user interface was not built to protect people who stray from the intended ways of using it.

If you choose to try any of what follows, you do so entirely at your own risk. Anything you conclude or implement from this book is your responsibility.

Ground rules if you decide to proceed

1. Never copy another person's settings. Your body is not theirs. What works for someone else can be dangerous for you. Investigate, test, and adjust based on *your* data.
2. Do not “learn by doing” with dangerous tools. Pressing buttons without understanding them, with a DIY closed-loop insulin system, is a serious risk.
3. Start disconnected. In your early testing phase, run the system as a dummy — not attached to your body — in parallel with your current glucose management, until you understand its behavior.

4. Stay connected to the community. Share experiences. The only way this work improves is through users who keep talking to each other and to the developers.

P R E F A C E

How to Use This Book



Who this book is for

This book is for people with Type 1 diabetes who already run a well-tuned hybrid closed loop (HCL) and want to explore going fully hands-off — letting the algorithm manage meals without any bolus, carb entry, or meal announcement from the user.

It was written for users of AAPS (AndroidAPS) with the autoISF development variant. Most of the content applies equally to users of Trio and iAPS (the iPhone-based oref variants), with some caveats noted where those systems lack certain features.

If you are not yet comfortably running a hybrid closed loop, or if you are still using Autotune or dynamicISF to paper over underlying profile problems, please pause here. Chapter 1 explains why that foundation matters so much. Without it, the rest of this book will not help you.

How the book is organized

The chapters follow the order you should work through them.

- Part I — Getting Oriented explains what autoISF Full Closed Loop is, how it differs from a hybrid loop, and whether it is a realistic goal for you.
- Part II — Foundations covers what must be true about your current system before you begin (Chapter 1), and a mental model of how autoISF actually works (Chapter 2, later).

- Part III — Core Setup walks through the settings you will change in AAPS Preferences (Chapter 3) and the heart of the project: tuning your ISF weights to your meal spectrum (Chapter 4).
- Part IV — Day-to-Day Operations shows how to nudge the loop when it needs nudging, and how to handle exercise.
- Part V — Alternative Modes describes partial or less-intensive paths to FCL if you cannot or do not want to commit to the full project.
- Part VI — Monitoring & Troubleshooting covers what to watch and how to diagnose problems.
- Part VII — Tools explains the Emulator, an optional but powerful way to replay and analyze your loop's decisions.

A note on the writing

Throughout the book, you will see callout boxes:

 **Warning** — A safety issue you must not skip.

 **Tip** — *A practical shortcut or lesson learned the hard way.*

 **Background** — *An aside for readers who want the reasoning behind a recommendation. Safe to skip on first reading.*

 **Example** — *The author's personal numbers, given for illustration. Do not copy them.*

When a section references another part of the book, it will say “(see Chapter X)” or “(see §X.Y)”. Cross-references are there to help you circle back; you are not expected to jump around on first reading.

PART I

Getting Oriented



What this is, whether it's for you, and what the journey looks like.

INTRODUCTION

What Is autoISF Full Closed Loop?



From hybrid to full closed loop

In a hybrid closed loop (HCL), the loop manages your basal insulin and delivers small correction boluses (SMBs) between meals, but *you* still bolus for meals. You tell the system you are about to eat; it does the rest.

In a full closed loop (FCL), you do nothing at mealtime. No bolus, no carb entry, no announcement. The loop watches your blood glucose, notices when it starts to rise, and delivers insulin on its own.

The attraction is obvious: for most people with T1D, meal management is the single biggest daily burden. Eliminating it — cleanly, safely, and with good time-in-range — is the prize.

Two approaches to FCL

The AAPS ecosystem now offers two ways to run a full closed loop:

1. FCL with Automations (AAPS Master). Available since autumn 2023. Simpler to set up. Uses personalized Automations to boost insulin delivery when a rise is detected. Covered in Chapter 8 (Other Avenues).

2. FCL with autoISF. The subject of this book. More sophisticated, more work to set up, capable of higher time-in-range and more daily “freedom” from intervention.

A clinical study of 16 AAPS users running the simpler Automations-based FCL showed an average of roughly 80% time-in-range without much tuning effort. That is a respectable outcome and a reasonable starting point if autoISF feels overwhelming.

autoISF aims higher — but it asks substantially more of the user up front.

Is autoISF FCL for you?

This is worth a moment of honest self-assessment.

autoISF FCL is probably a good fit if:

- You have a well-tuned hybrid closed loop already — your profile ISFs, basal, and IC are experimentally correct, not “covered up” by Autotune or dynamicISF.
- You use Lyumjev or Fiasp (or, with caveats, Apidra).
- You use a Dexcom G6 or another CGM with reliably smooth values.
- You have several weeks of “free head” to invest in set-up.
- You enjoy reading, analyzing, and iterating on your own data.
- You are comfortable with the idea that some meals will not be perfect, and you are not chasing an HbA1c under 5.5%.

autoISF FCL is probably not for you if:

- Your hybrid loop is held together by many custom Automations or “tricks.”
- You use an insulin slower than Fiasp.
- Your CGM is jumpy or you have unreliable Bluetooth connectivity.
- You are chasing very tight time-in-range (time above 140 mg/dL < 10%) — that level of control generally still requires hybrid looping with meticulous pre-boluses, and often low-carb diet and/or a GLP-1 co-medication.
- You are looking for something you can set up in a weekend and then forget.
- You prefer to proceed mostly by trial and error.



A REALITY CHECK ABOUT INDUSTRY “FCL” CLAIMS

You will see headlines about commercial systems achieving “full closed loop” results. Read the fine print. Most of those studies enroll patients with HbA1c in the 7–9% range and simply show that the commercial FCL is “no worse” than the patient’s previous HCL. That is a meaningful advance for that population. It is not what this book is about. DIY FCL plays in a different league — it demands more from you, and it can deliver more.

A different goal, a different mindset

Going into this project, please internalize this idea:

With 18 tuneable parameters, you can always “trick” your loop into handling one meal beautifully. But settings that handle one meal well are not the same as settings that handle all your meals well.

Every chapter that follows is in service of the second goal. The author’s experience is that it is entirely possible to find a single set of settings that gets you through your normal meal spectrum with good time-in-range. It just takes discipline to get there.

If autoISF is too much right now

Several gentler on-ramps exist:

- AAPS Master with FCL Automations — see Chapter 8.
- Meal Announcement mode — keep giving a small pre-bolus, but otherwise let autoISF do the work. See Chapter 7.
- Other dev variants — Boost, AIMI, EatingNow, Tsunami. Each trades off differently. Briefly surveyed in Chapter 8.

There is no shame in picking a simpler path. The best loop is the one you can actually run.

A note for iPhone users (Trio and iAPS)

autoISF has been ported into rapidly evolving development branches of Trio and iAPS for iPhone. The algorithm is the same, but Trio and iAPS users are at some disadvantage for FCL because:

- They lack Automations (Trio’s “Shortcuts” have limited options).
- There is no e-book variant written specifically for their systems.
- Many users did not go through the AAPS “Objectives” tutorial sequence, and may not have the solid vanilla HCL starting point this book assumes.

For Trio users, general documentation is still being written. If you are a Trio user, the best plan is to get your HCL rock-solid first — by the time you are ready for FCL, there will probably be more Trio-specific material.

iOS Loop users: there is no comparable FCL option, because the iOS Loop algorithm depends heavily on carb inputs.

Where to find the software

- autoISF quick guide and ongoing development: <https://github.com/ga-zelle/autoISF>
- AAPS build with autoISF: <https://github.com/T-o-b-i-a-s/AndroidAPS/> (use the highest available version)
- Trio with autoISF (TAI): <https://github.com/mountrcg/Tai>
- iAPS with autoISF: <https://github.com/mountrcg/iAPS>
- Discord community: <https://discord.gg/tamvhh57Xs>

Always use the most recent branch compatible with your AAPS version. Note that the newest autoISF sometimes lags the newest AAPS Master by a few weeks; if you are migrating, read the current Discord threads before upgrading.

The Journey to Full Closed Loop



This chapter gives you the whole journey in one view before you start digging into the details. Read it once, come back to it as a checkpoint, and use it to decide where you are in the project.

The big picture: four phases

Phase 1 – Make sure you're ready	(weeks -2 to 0)
Phase 2 – Configure safety settings	(week 1)
Phase 3 – Tune ISF weights to meals	(weeks 2-6)
Phase 4 – Handle exceptions	(weeks 6+)

Total realistic timeline: 4–8 weeks of actively working on it, on top of your normal diabetes management. Some people take longer. Nobody does it overnight.

Phase 1 — Make sure you're ready (Chapter 1)

Before you touch a single autoISF setting, confirm all of the following are true:

- ☐ Your hybrid closed loop is tuned and performing satisfactorily.

- ☐ Your profile ISFs, basal, and IC are experimentally correct (not derived from Autotune).
- ☐ You are not relying on dynamicISF to cover up an incorrect profile.
- ☐ You use Lyumjev or Fiasp.
- ☐ Your CGM values are smooth and reliable (ideally Dexcom G6, overlapped).
- ☐ Your pump/cannula/pod setup delivers reliably, without frequent occlusions.
- ☐ Your Bluetooth connectivity is stable 24/7.
- ☐ You have a few weeks of mental space to work on this.

If any of these is shaky, fix it before you begin. The rest of the project will either fail, or it will paper over the weakness in ways that hurt you later.

Chapter 1 goes into each of these in depth.

Phase 2 — Configure safety settings (Chapter 3)

You cannot run FCL on HCL-era safety restrictions. In HCL, the user gives big boluses for meals, so the loop never needs to deliver big SMBs. In FCL, the loop must deliver big SMBs — because you are not bolusing.

So first, you widen the guardrails, and then you install new ones.

Specifically, in this phase you will:

1. Widen the max allowed SMB size (SMB range extension).
2. Widen the max allowed ISF amplification (autoISF_max).
3. Nudge up the SMB delivery ratio.
4. Install your iob threshold (iobTH) — this is your most important new safety setting.
5. Configure a handful of other AAPS preferences (Autosens off, exercise settings, odd/even target logic, etc.).

You will not enable “ISF adaptation by glucose behavior” yet. That is Phase 3.

Chapter 3 walks through each setting.

Phase 3 — Tune ISF weights to your meals (Chapter 4)

This is the heart of the project, and where most people spend the most time.

autoISF has five ISF “weight” parameters that each apply during a different phase of the post-meal glucose curve:

<i>Parameter</i>	<i>When it matters</i>
bgAccel_ISF_weight	The first signs of a rise (acceleration)
pp_ISF_weight	The steep linear rise after a big meal
bgBrake_ISF_weight	The curve flattening toward the peak
dura_ISF_weight	Stuck-high plateaus, especially fatty meals
bg_ISF_weight	Very high bg levels

You tune them in that order, starting with bgAccel_ISF.

The rule of thumb is: within ~20 minutes of a rise being detected, the loop should have delivered about the same amount of insulin you would have bolused in HCL for that meal. That is the benchmark you’re calibrating against.

You iterate. You look at two or three meals from your normal spectrum. You check the AUTO ISF tab (or use the Emulator — Chapter 11 — to make this easier). You adjust, watch another day or two of meals, adjust again.



THE MOST IMPORTANT DISCIPLINE IN PHASE 3

Work through meals with your normal insulin sensitivity only. Do not try to tune during an illness, a sports day, a hormonal shift, or anything else that changes your insulin need. You are trying to build a baseline that the rest of the system adapts from.

Chapter 4 walks through each weight, how to start, and how to iterate.

Phase 4 — Handle exceptions (Chapters 5 and 6)

Once your meal management is working, you turn to everything else:

- Chapter 5 — Modulating aggressiveness. Automations that switch the loop between “standard,” “milder,” and “more aggressive” modes. Odd/even target tricks for emergency SMB shutoff. The concept of a “DIY cockpit” — custom buttons on your AAPS home screen for recurring situations (snacks, kids’ meals, etc.).
- Chapter 6 — Exercise and activity. Dynamic iobTH, the exercise mode, the activity monitor (step-counter-driven sensitivity adjustment), and the particularly tricky “exercise after a meal” scenario.

Most users do not need everything in these chapters. Pick what maps to your actual life.

A staged implementation strategy

You do not have to do everything at once. Many people start by running FCL only at:

- A single meal per day (e.g., only dinners).
- Weekdays only.
- A specific time window, with a fallback to HCL otherwise.

This is a very good idea in your first few weeks. You can expand once you see the loop behave the way you expect.

What success looks like

When the project is done, you should see:

- Most meals handled hands-off, with glucose staying below ~180 mg/dL.
- Iob rising quickly after meal start (within 10–20 minutes) and then capping at your iobTH.
- Occasional need for a small snack to catch an insulin “tail” — this is normal.
- Very few manual interventions per day.
- A rolling weekly time-in-range you are happy with — for most adults on a normal diet, somewhere in the 75–90% range.

You are *not* trying to never go above 140 mg/dL. If that is your goal, FCL is the wrong path. See the discussion in Chapter 1.

Safety rail: how to exit FCL

You can always go back to HCL. Two ways:

1. AAPS Preferences → OpenAPS SMB → autoISF → “Enable ISF adaptation by glucose behavior” OFF. This disables autoISF entirely. You are now running vanilla OpenAPS SMB+UAM.
2. Tap the violet FCL loop icon on the AAPS home screen and pick the green HCL icon (if your version supports it).

You will need to re-enable the Insulin button (AAPS Preferences → Overview → Buttons) to bolus for meals again.

Keep this escape route in mind throughout. If something feels wrong, go back to HCL. Always.

A note on case studies

The original document refers frequently to numbered “case studies” (e.g., “Case Study 4.1: Pizza,” “Case Study 6.2: Biking day with hi-carb lunch”). These sit alongside the main text on the author’s GitHub. As you work through this book, look up the case study for any situation that feels close to your own. They are the most concrete material available and will save you a lot of reasoning from first principles.

A full index of case studies is in Appendix B.

PART II

Foundations



*What must be true before you start, and how the algorithm
actually works.*

Prerequisites for Full Closed Loop



Everything in this chapter is about what must be true *before* you begin. If any of these foundations is missing, FCL will either fail outright or fail quietly by trading off things you did not realize you were trading off.

⚠ The most common way this project fails is not a tuning error. It is someone who skipped a prerequisite, told themselves they could work around it, and spent six weeks chasing a symptom whose cause was a weak foundation.

1.1 A well-tuned hybrid closed loop

Your first prerequisite is a hybrid closed loop that is already performing satisfactorily. That means:

- You have experimentally determined, correct profile ISFs for each hour of the day (not ISFs produced by Autotune or implied by dynamicISF).
- Your basal holds you steady in open-loop testing.
- Your insulin-to-carb ratio (IC) has been validated against real meals.
- Your DIA is set correctly for your insulin.

- Your hybrid closed loop allows SMB sizes up to 120 minutes of basal without trouble.

There are three reasons this matters so much:

1. autoISF is highly personalized. The loop will be asked to mimic *your* successful HCL behavior. If your HCL behavior was the result of errors that happened to cancel out, autoISF will inherit and amplify those errors.
2. autoISF multiplies your profile_ISF. Every tuning parameter you will set in Chapter 4 is a *factor on your profile ISF for that hour*. If your profile ISF is wrong by 20%, every bgAccel, pp, and dura weight you tune will be calibrated to compensate — and will be wrong for the same reason.
3. New parameters stack on old ones. If you tune new parameters on top of an unfixed old profile, you create a system of balanced errors. This can work in individual scenarios but creates an unstable system overall, and is nearly impossible to untangle later.



IF YOU'VE BEEN RUNNING WITH DYNAMICISF, AUTOTUNE, OR A MAZE OF
CUSTOM AUTOMATIONS

You may have “learned to live” with a profile that is not quite right, because the dynamic components were hiding the problem. Those components will now be turned off (for FCL, you will also turn off Autosens). The underlying problem will surface. Fix it in HCL first, or your FCL project will produce confusing, unstable results.

For help getting your HCL foundation right, see the neighboring HCL repository: <https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings>

1.2 *A fast insulin*

FCL only works with a fast insulin. Period.

In FCL, you are no longer pre-bolusing. The loop must recognize that a rise has started and deliver insulin fast enough to meaningfully limit the peak. With a slow insulin, by the time the insulin is active, the peak has already happened and passed.

Use Lyumjev or Fiasp.

Possible exceptions:

- Apidra may work — see Case Study 1.2 — though with a lower margin for error.
- Humalog does not work well with a normal-carb diet in FCL.
- If you are on a consistent low-carb diet, are on a GLP-1 drug, or have gastroparesis, a slower insulin may be acceptable. All three situations slow down your carb absorption enough to match a slower insulin.

A modelling study compared insulins in an FCL context and found that faster insulins:

- Produce significantly lower glucose peaks.
- Tolerate a few minutes of delay in detection without catastrophic peaks.
- Minimize the effect of meal size variability on peak height.

For deeper reading: *The artificial pancreas and meal control...* (available as a PDF in the HCL repo linked above).

1.3 Reliable insulin delivery

Your pump, cannula, and insulin must all cooperate reliably with your chosen fast insulin.

INSULIN TOLERANCE

Occlusions kill FCL performance faster than almost anything else. If your pump and cannula do not tolerate Lyumjev (or Fiasp) well — if you see hard-to-explain glucose rises at ever-increasing “fake” iob before a scheduled cannula change — FCL will not work for you.



CANNULA/POD TIMING

Many users find that 48 hours is the practical limit for their cannulas, well before any manufacturer-stated change interval. Change early. You can easily lose 25% time-in-range on a day where a site is starting to fail (see Case Study 1.1).



**IF YOU ARE COMING FROM LEAKING PODS OR ERRATIC SENSITIVITY SWINGS
THAT YOU “KIND OF CONTROLLED” WITH DYNAMICISF**

Do not start FCL. Resolve the pod/cannula problem first. FCL has no tolerance for erratic delivery.

BLUETOOTH CONNECTIVITY

In FCL, the loop must deliver the right SMB at the right moment. If the pump loses Bluetooth connection during the first five minutes of a meal, you have lost the most important window for insulin delivery.

- In AAPS Preferences → Local Alerts, turn on the connection-lost alert.
- Consider an Automation that fires an alarm if Bluetooth has been down for more than a few minutes.
- Keep your phone close to the pump 24/7.

See Case Studies 1.4 and 1.6 for what goes wrong when this prerequisite fails.

1.4 *An excellent CGM*

Because you are no longer giving a meal bolus, the CGM’s job becomes dramatically more important. The loop is reading glucose values and deciding, in real time, “is this the start of a meal?” A jumpy CGM will produce false “yes” decisions.

Things to understand about your CGM, before starting FCL:

1. Does it support “SMB always”? You need SMBs to be deliverable even when $cob = 0$ (which is *always the case* in FCL).
2. What smoothing, if any, is applied? See <https://androidaps.readthedocs.io/en/latest/Usage/Smoothing-Blood-Glucose-Data.html>
3. Does your CGM produce artefacts? Review several days of your own data, looking for sudden jumps that the loop could mistake for a meal. Compression lows at night are a particularly common culprit.

4. If your CGM requires calibrations, calibration events often produce jumps. You will need a protocol for pausing FCL around calibrations.

STANDARD 5-MINUTE CGMS

The best-proven setup, as of this writing, is Dexcom G6 with overlapping sensors on each arm (Case Study 1.5). Overlap guarantees you always have at least one reliable sensor running.

G6 production is scheduled to end mid-2026. Alternatives are being evaluated; at time of writing, Syai Ultra (Syai Health, Singapore) and CareSens Air (i-Sens, South Korea) look promising based on a 10-CGM comparison by one autoISF FCL user. Watch the community for more data.

To ease the transition from G6 to a newer sensor:

- Consider stockpiling a few G6 sensors (via the Anubis extension, or from users who are switching).
- When you start your new sensor system, alternate: every second sensor is your G6. This lets you compare relative performance and build a stock of the new sensor.
- Share your findings! Case Study 1.5 needs updating with mixed-sensor data.



AUTOISF'S BUILT-IN JUMP DETECTOR

autoISF automatically blocks SMBs if the delta between two consecutive 5-minute CGM values exceeds 30% of the current bg (or 20% at targets above 100 mg/dL). This protects against artefacts. The side effect is that a genuine steep rise — for example, after a large sweet drink — can also briefly block SMBs. For 5 minutes, you lose the opportunity for a large early SMB; the loop will catch up in subsequent decisions, but a bit of peak height is lost. This is one reason to think carefully about sweet drinks at meal start.

1-MINUTE CGMS

autoISF has supported 1-minute CGM values since version 3.0.1. The data flow is Libre → Juggluco → AAPS, with exponential smoothing applied in AAPS.

Early results show smoother, more frequent, smaller SMBs. Whether the 1-minute sampling actually produces *earlier* acceleration detection is still being evaluated. For details, see the “Using 1-minute data in autoISF” chapter of the autoISF Quick Guide.

Some implications for later chapters:

- SMB range extension (Chapter 3) can usually stay at default 1.0.
- SMB delivery ratio (Chapter 3) should typically be under 0.5, not boosted.
- Logfile retention on the phone is shorter, so copy logs to your PC more often.

1.5 Meal-related limitations

There are essentially no meal limitations for FCL, as long as you use a fast insulin.

A few nuances:

- Fat- and protein-rich meals are easier, not harder. Late carb release covers the insulin tail.
- Gastroparesis or GLP-1 drugs also make FCL easier by slowing absorption.
- Erratic consumption of small, fast-carb snacks or sweet drinks is the main challenge. This can be handled via a “DIY cockpit” button (Chapter 5, §5.4) with a couple of keystrokes, but it is a deviation from the pure hands-off ideal.

You do not need a low-carb diet to run FCL. You do need to decide how much “nudging” you are willing to do for snacks and sweet drinks, and how much time-in-range you are willing to sacrifice if you don’t.

1.6 Lifestyle-related considerations

TECHNICAL RELIABILITY

FCL requires 24/7 reliability of:

- CGM signals
- Pump Bluetooth
- Phone proximity to both
- Cannula/pod performance

Depending on your lifestyle, this is either routine or burdensome. Be honest with yourself. If you regularly forget your phone, leave your pump on a table across the room, or let cannulas run to 72 hours hoping for the best — this project will frustrate you.

EXERCISE PREPARATION

In HCL, preparing for exercise means reducing iob ahead of time. In FCL, the algorithm is tuned to *raise* iob aggressively when a rise is detected. So setting a pre-exercise high temporary target and reduced %profile, effective *before* the meal, would sabotage your meal management.

Exercise management in FCL therefore needs its own strategy. Chapter 6 covers:

- Dynamic iobTH (the exercise mode’s ability to auto-lower the iob cap).
- The activity monitor (step-counter-driven small adjustments).
- A “DIY cockpit” button for the exercise-after-meal scenario.

Unusual activity requires disciplined preparation — especially if you want to avoid mid-exercise snacking.

CHILDREN

FCL for children presents extra challenges:

- Lyumjev may not be available or tolerated.
- Hourly basal rates are low, limiting SMB size.
- Diets often include fast carbs that produce steep spikes, made worse by small body volume.
- Sensitivity shifts and circadian changes happen more sharply.
- Bluetooth discipline and cannula site discipline are harder to enforce.
- Caregiver-child coordination adds complexity.

See Chapter 7 (Advanced HCL with Meal Announcement) for a gentler intermediary approach, which is often more practical for children.

1.7 Time to set up

Set up is a real project of real weeks. The exact timeline depends on:

- Your current HCL tuning quality.
- How varied your meals and lifestyle are.
- How much compromise you will accept (cannula changes more often, skipping meals when glucose is high, etc.).

- Your personal aptitude for data analysis and iteration.

THE STRUCTURED PATH

Follow the chapter sequence. Do not shortcut.

But: you do not have to implement everything at once. Many users start by:

- Running FCL only for one meal type per day (dinners, for example).
- Running FCL only on weekends, with HCL during the work week.
- Finding feasible settings for the meal they most care about first, then expanding.

Once one meal is working well, expand to others, then turn to exercise and the other non-meal challenges in Chapters 7 and 8.

THE “TRIAL AND ERROR” FAST TRACK

You will find FCL success stories from users who jumped in, added patches as problems arose, and never did the structured setup. Some of these solutions are good enough for the person who built them. But they tend to be:

- Unstable
- Poorly understood by their own author
- Nearly impossible for anyone to help debug
- Impossible to fine-tune later without starting over

If you go this route, be prepared to start over at some point. And recognize that as your own web of Automations and workarounds grows, no one in the community will be able to help you untangle it.



A WORD FROM THE AUTHOR

“Necessity is the mother of invention.” Creative users regularly come up with solutions that push the whole community forward. I welcome that spirit. Just do it with your eyes open — expect some restarts, and document what you do so others can learn from it.

SAFETY IS NON-NEGOTIABLE

No matter how you approach FCL, these safeguards are mandatory:

1. Set an iob threshold (iobTH) above which no more SMBs will fire. This is a built-in feature of autoISF (Chapter 3, §3.4). For FCL methods that do not have a built-in iobTH, set up an Automation to mimic one.
2. Know how to shut off SMBs quickly. Setting an odd-numbered temporary target blocks all SMBs while active. This is your emergency brake. Use it for:
 - Anti-hypo snacks
 - Unusual situations where you do not want the loop to react aggressively
 - Any moment when something feels off
3. Practice exiting FCL. In your setup weeks, deliberately switch from FCL to HCL and back several times, so you know the motions by heart before you ever need them in a hurry. See Chapter 10 (Troubleshooting, §10.1) and Chapter 5 (§5.6).

1.8 Summary checklist

Before you move on to Chapter 2, confirm:

- ☐ My hybrid closed loop is satisfying, without relying on Autotune or dynamicISF.
- ☐ My profile ISFs, basal, and IC are experimentally validated.
- ☐ I am using Lyumjev or Fiasp.
- ☐ My pump, cannula, and insulin tolerate each other. Occlusions are rare.
- ☐ My CGM is smooth and reliable. I have a plan for calibration jumps.
- ☐ My Bluetooth connectivity is stable 24/7.
- ☐ I have 4–8 weeks of mental bandwidth for this project.
- ☐ I have identified a subset of meals to start with (e.g., dinners only).
- ☐ I know how to exit FCL back to HCL.
- ☐ I have joined the community (Discord link in the Introduction).

When every box is checked, you're ready for Chapter 2.

How autoISF Works



Before you touch any settings, you need a mental model of what autoISF actually does. This chapter is for the “why” behind the “how.”

2.1 The core idea

Every 5 minutes (every minute, if you’re on a 1-minute CGM), autoISF does this:

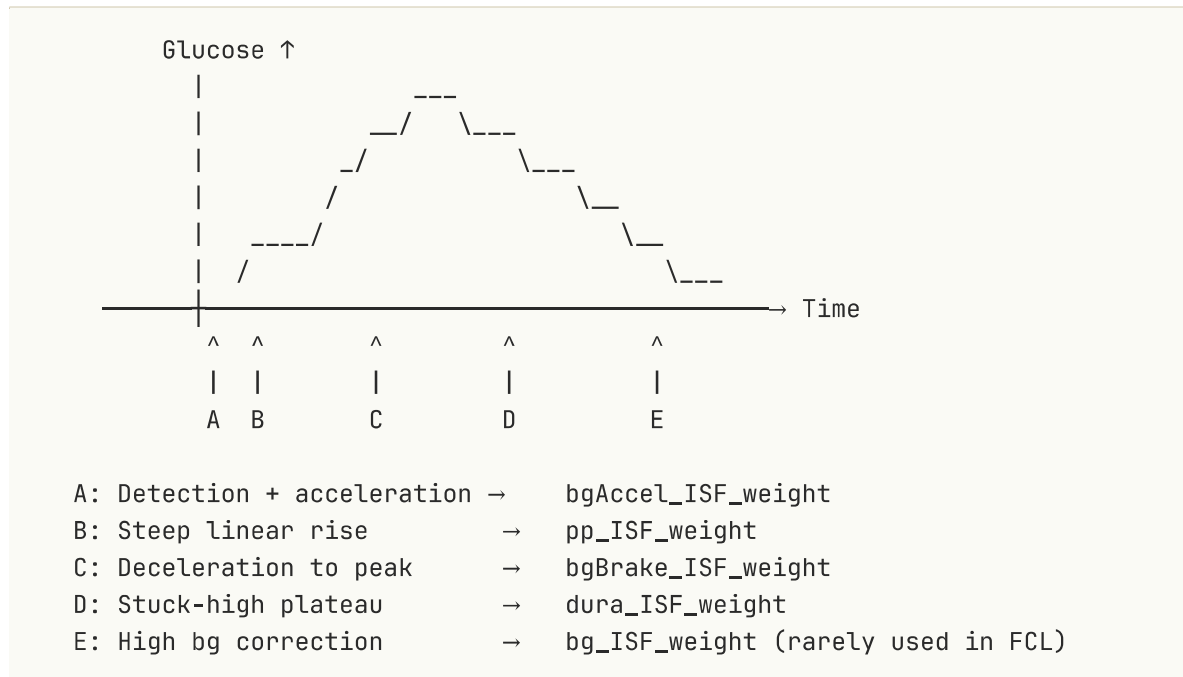
1. Reads the recent glucose curve — last several CGM values, smoothed, with a parabola fitted to detect acceleration.
2. Classifies the phase — is this an acceleration? A steep linear rise? A deceleration? A plateau?
3. Calculates an adapted ISF (called **sens** in the AUTO ISF tab) by applying modulation factors to your `profile_ISF` for that hour.
4. Asks the standard `oref(1)` UAM loop what insulin is needed given that adapted ISF.
5. Delivers an SMB and/or adjusts temp basal, capped by several safety limits.

The key insight: your `profile_ISF` is the anchor. autoISF does not replace it. It multiplies it up (to deliver more insulin) or down (to deliver less), within bounds you set.

This is why Chapter 1 insisted on a correct `profile_ISF`. Every single thing in the rest of this book is a factor on that number.

2.2 The five weight parameters, mapped to the glucose curve

The most important consequence of autoISF’s design is that different phases of a meal’s glucose curve get different ISF modulation, each controlled by its own tuneable weight:



In FCL, the first two (bgAccel and pp) do most of the work. They are the “initial iob ramp-up” that replaces your former meal bolus. The others play supporting roles for specific meal types.



ORIGIN STORY (SKIPPABLE)

The lead developer, ga-zelle, arrived at these parameters by analogy rather than by reading endocrinology papers. `bgAccel_ISF` came from Newton’s second law — if glucose is being “pushed up,” the resistance (ISF strength) should scale with the push. `dura_ISF` came from control theory — a PID controller has a proportional, integral, and derivative term; the integral was missing from oref, so `dura_ISF` adds it. He later realized the body’s own insulin response also has a fast initial phase and a slower sustained phase, which is roughly what the two mechanisms recreate. Interesting, not essential.

2.3 The guardrails

autoISF’s aggressive ISF modulation is hemmed in by several settings you will configure in Chapter 3:

<i>Guardrail</i>	<i>What it limits</i>
autoISF_max	Maximum amplification factor on your profile_ISF (default 2.0).
autoISF_min	Minimum factor — how soft the loop can go (default 0.3 in this book).
SMB range ext.	Max SMB size as a multiple of 2 hours of basal.
SMB delivery ratio	Fraction of calculated insulin need delivered via each SMB.
iobTH	Absolute ceiling — once iob exceeds this, SMBs stop firing.
max iob	Hard ceiling, separate from iobTH.
30% jump rule	SMBs auto-blocked if the CGM delta exceeds 30% of current bg.
Odd-numbered target	Shuts off SMBs entirely. Your emergency brake.

iobTH is the most important one. It is what separates “a loop that sometimes delivers 2U because the glucose curve looked ambiguous” from “a loop that kept firing SMBs until 12U of iob had accumulated.” You will set it in Chapter 3.

2.4 Two safety mechanisms that people trip over

EVEN/ODD TARGET RULE

autoISF uses a clever trick to let you shut SMBs on and off with a single keystroke:

- Even-numbered glucose target (profile or TT) → SMBs allowed, as normal.
- Odd-numbered glucose target → SMBs blocked. The loop still adjusts via temp basal, but won’t deliver an SMB.

For mmol/L users, the logic uses the first decimal — e.g., 5.0 is “even,” 5.1 is “odd.”

This feature is enabled in AAPS Preferences → OpenAPS SMB → autoISF settings → “Enable alternative activation of SMB depending on current target” = ON.

In FCL you will keep this permanently on. You will use odd-numbered targets as your emergency brake (for anti-hypo snacks, compression lows, “I want a moment to think,” etc.).

30% JUMP RULE

If the delta between two consecutive CGM values exceeds 30% of the current bg (20% at targets above 100 mg/dL), SMBs are automatically blocked for that cycle.

Why this matters: on a well-behaved CGM this almost never triggers for real meals. On a jumpy CGM, or with a sweet drink that causes a genuine sharp spike, it can block the very SMB you wanted.

Example: From 74 mg/dL, a jump to 97 mg/dL (+23, which is +31% of 74) blocks SMBs. That’s 5 lost minutes of insulin delivery at the worst possible moment.

Mitigations, in preference order: 1. Use a CGM that doesn’t produce artefacts (Dexcom G6 overlapped is the gold standard). 2. Avoid sweet drinks at meal start. 3. Review logs to see how often this triggers for you, and adjust behavior.

Do not change the 30% threshold in the code.

2.5 Where to look when you want to know what happened

Three places give you the information you need:

1. The AUTO ISF tab in AAPS. At any moment, shows the most recent loop decision in detail — profile_ISF, each modulation factor, the final sens, any limits that were hit, the calculated SMB size, and whether it was delivered or blocked and why.
2. Extra graphs below the AAPS home screen glucose chart. These can display iobTH, the autoISF factor contributions over time, the fitted parabola, and more. Enable them from the chart options.
3. The Emulator (Chapter 11–14). Replays your logfiles on a PC or phone and shows loop decisions in tabular form, plus lets you ask “what if I had used setting X instead?”

For setup and tuning, you will consult the AUTO ISF tab constantly. For deeper analysis, the Emulator is worth installing.

2.6 *What autoISF does not do*

A few common misconceptions:

- autoISF does not replace Autosens or dynamicISF. It serves a different purpose — reacting to the glucose curve in real time, not tracking longer-term sensitivity trends. For FCL, you will turn both Autosens and dynamicISF off.
- autoISF does not “learn” from your meals over time. It reads the current curve, applies your settings, and decides. Tuning is your job.
- autoISF is not a carb-counting replacement. It reacts to the curve, not to a meal. If you take a wildly out-of-spectrum meal, it will over- or under-treat, and you will need the manual levers in Chapter 5.
- autoISF is not a bolus calculator. It delivers many small SMBs based on predicted insulin need over the next 30–60 minutes, recalculated every 5 minutes. If you give a user bolus on top, you distort the curve autoISF is reading, and the predictions drift.

With this model in mind, let’s start configuring.